

Hari Mohan Prajapat

Bangalore, India

Applied ML / Search Engineer – LLM & Relevance Systems

+91 78923 39693

✉ softengineerhari@gmail.com

🐙 GitHub | 🔗 LinkedIn

SUMMARY

Applied ML Engineer with **6.8 years of experience** building production-grade systems spanning **search, retrieval, ranking, and LLM-powered Q&A**. Strong background in **RAG pipelines, LLM integration, query understanding, and agentic workflows**, combined with deep systems engineering expertise in **distributed APIs, evaluation loops, and cloud-native deployment**. Experienced in iterative quality improvement (hillclimbing) and shipping reliable, scalable user-facing AI systems.

CORE EXPERTISE

- **Search & Relevance:** Retrieval pipelines, query understanding, ranking signals, relevance evaluation
- **LLMs & GenAI:** Claude, LLaMA, LiteLLM, prompt tuning, RAG, agentic workflows
- **Machine Learning:** Model training, evaluation, feature engineering, TensorFlow, PyTorch, scikit-learn
- **Backend Systems:** Python, FastAPI, Flask, Django, REST APIs, Microservices, Background Workers
- **Data & Storage:** PostgreSQL, MySQL, AWS Aurora, MongoDB, Cassandra, structured & unstructured data
- **Vector Databases:** Pinecone, Milvus, Weaviate
- **Data Pipelines & Streaming:** Apache Kafka, Apache Airflow, Apache Spark
- **Cloud & Platform:** AWS (Lambda, EC2, S3, API Gateway), Azure, Google Cloud Platform (GCP)
- **Containerization & Orchestration:** Docker, Kubernetes
- **Evaluation & Quality:** Offline evaluation loops, iterative improvement, monitoring
- **CI/CD & DevOps:** Git, Jenkins, GitHub Actions, GitLab CI/CD, observability
- **Frontend (Supporting):** React, Next.js, TypeScript

PROFESSIONAL EXPERIENCE

- **UST Global (Merck, Deluxe)** 2024 – 2025
Senior Software Engineer – Search & AI Platforms Bangalore, India
 - Led design and implementation of **LLM-powered search and Q&A systems** for enterprise knowledge discovery.
 - Built **RAG retrieval pipelines** using LangChain and LlamaIndex; optimized chunking, retrieval, and prompts to improve relevance.
 - Integrated **Claude and LLaMA via LiteLLM**, enabling controlled experimentation and cost-aware inference.
 - Designed **agent-style workflows** (AutoGen-inspired) for multi-step query understanding and response synthesis.
 - Implemented **iterative evaluation loops** supporting continuous quality hillclimbing.
 - Owned **production ML workflows** on AWS (Lambda, EC2, S3, API Gateway) with monitoring and rollback.
 - Partnered with product and platform teams to align relevance goals and accelerate delivery.
- **Genpact (Morgan Stanley)** 2024
Software Engineer – Applied ML & Retrieval Systems Bangalore, India
 - Built **LLM-powered assistants** for financial data exploration, summarization, and Q&A.
 - Designed **retrieval workflows** over structured and unstructured datasets using LangChain.

- Experimented with **agentic AI patterns** for query decomposition and multi-step analytics.
- Ensured **trustworthy and auditable AI outputs** in regulated environments.
- Deployed high-throughput backend services using Python FastAPI, PostgreSQL, and AWS.
- **Atos Syntel (PayPal, State Street)** 2022 – 2023
Software Engineer Bangalore, India
 - Built scalable backend services and React dashboards for financial platforms.
 - Improved application performance by **55–65%** via API and rendering optimizations.
- **Altisource (Hubzu Auctions)** 2021 – 2022
Software Engineer Bangalore, India
 - Developed high-traffic, real-time UI systems for live auction platforms.
 - Improved system stability during peak traffic events.
- *Earlier Experience: Mercedes-Benz (ATCS), Moonraft, Medi Assist (2018–2021)*

PROJECTS

- **Enterprise RAG Platform** — End-to-end retrieval, ranking, and LLM-based Q&A systems
- **ML Learning Platform** — Model training and experimentation using Python and Jupyter
- **SDLC Tracker (MERN)** — Engineering metrics and delivery insights platform

EDUCATION

- **Bachelor of Technology — Information Technology**
Rajasthan Technical University, Kota

AWARDS

- **UST Global Best Frontend Developer** — 2025
- **Genpact Best Team Player** — 2024